

Microproject 01

PRN: 25UAM095

Division: B

Title: Online shopping cart-Each node stores a cart item — product ID, name, unit price, and quantity.

Doubly linked so the user can scroll forward and backward through cart items on a device.

Operations to implement

- Add item: append new node; if item exists, increment quantity instead
- Remove item by product ID: unlink node, fix both prev and next pointers
- Update quantity: traverse to node, modify qty; delete if qty becomes 0
- Compute bill: traverse forward, accumulate price × qty; apply 10% coupon if applicable
- Display cart in reverse (tail to head) using *prev — "most recently added first"

Code:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
struct Node {  
    int productID;  
    char name[50];  
    float price;  
    int quantity;  
    struct Node *prev;  
    struct Node *next;  
};
```

```
struct Node *head = NULL;

struct Node *tail = NULL;


// Clears full line - fixes your "rs" problem
void clearInput() {
    int c;
    while ((c = getchar()) != '\n' && c != EOF);
}


void addItem() {
    int id, qty;
    char name[50];
    float price;

    printf("Enter Product ID: ");
    scanf("%d", &id);
    clearInput();

    struct Node *temp = head;
    while (temp != NULL) {
        if (temp->productID == id) {
            printf("Product already exists.\n");
            printf("Enter quantity to add: ");
            scanf("%d", &qty);
            clearInput();
            temp->quantity += qty;
            printf("Quantity updated.\n");
            return;
        }
    }
}
```

```
}
```

```
temp = temp->next;
```

```
}
```

```
printf("Enter Product Name: ");
```

```
fgets(name, 50, stdin);
```

```
name[strcspn(name, "\n")] = '\0';
```

```
printf("Enter Unit Price: ");
```

```
scanf("%f", &price);
```

```
clearInput();
```

```
printf("Enter Quantity: ");
```

```
scanf("%d", &qty);
```

```
clearInput();
```

```
struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
```

```
newNode->productID = id;
```

```
strcpy(newNode->name, name);
```

```
newNode->price = price;
```

```
newNode->quantity = qty;
```

```
newNode->prev = NULL;
```

```
newNode->next = NULL;
```

```
if (head == NULL) {
```

```
    head = tail = newNode;
```

```
} else {
```

```
    tail->next = newNode;
```

```
    newNode->prev = tail;

    tail = newNode;

}

printf("Product added to cart.\n");

}
```

```
void deleteItem() {

    int id;

    printf("\nEnter Product ID to delete: ");

    scanf("%d", &id);

    clearInput();


    struct Node *temp = head;

    while (temp!= NULL) {

        if (temp->productID == id) {

            if (temp->prev) temp->prev->next = temp->next;

            else head = temp->next;

            if (temp->next) temp->next->prev = temp->prev;

            else tail = temp->prev;

            free(temp);

            printf("Product deleted.\n");

            return;

        }

        temp = temp->next;

    }

    printf("Product not found.\n");

}
```

```
void updateQuantity() {  
    int id, qty;  
    printf("\nEnter Product ID to update: ");  
    scanf("%d", &id);  
    clearInput();  
    printf("Enter new Quantity: ");  
    scanf("%d", &qty);  
    clearInput();
```

```
  
    struct Node *temp = head;  
    while (temp!= NULL) {  
        if (temp->productID == id) {  
            temp->quantity = qty;  
            printf("Quantity updated.\n");  
            return;  
        }  
        temp = temp->next;  
    }  
    printf("Product not found.\n");  
}
```

```
  
void displayCart() {  
    if (head == NULL) { printf("\nCart is empty.\n"); return; }  
    printf("\nCart Items:\n");  
    printf("ID\tName\t\tPrice\tQty\tTotal\n");  
    struct Node *temp = head;  
    while (temp!= NULL) {
```

```

        printf("%d\t%s\t\t%.2f\t%d\t%.2f\n", temp->productID, temp->name, temp->price,
temp->quantity, temp->price * temp->quantity);

        temp = temp->next;

    }

}

```

```

void computeBill() {

    float total = 0;

    struct Node *temp = head;

    while (temp!= NULL) { total += temp->price * temp->quantity; temp = temp->next; }

    printf("\nTotal Bill Amount: %.2f\n", total);

}

```

```

void displayReverse() {

    if (tail == NULL) { printf("\nCart is empty.\n"); return; }

    printf("\nCart Items in Reverse:\n");

    struct Node *temp = tail;

    while (temp!= NULL) {

        printf("%d\t%s\t\t%.2f\t%d\t%.2f\n", temp->productID, temp->name, temp->price,
temp->quantity, temp->price * temp->quantity);

        temp = temp->prev;

    }

}

```

```

int main() {

    int choice;

    while (1) {

        printf("\n===== Shopping Cart Menu =====\n");

```

```
    printf("1. Add Item\n2. Delete Item\n3. Update Quantity\n4. Display Cart\n5.  
Compute Bill\n6. Display Cart in Reverse\n7. Exit\n");
```

```
    printf("=====\n");
```

```
    printf("Enter your choice: ");
```

```
    if (scanf("%d", &choice) != 1) { clearInput(); continue; }
```

```
    clearInput();
```

```
    switch (choice) {
```

```
        case 1: addItem(); break;
```

```
        case 2: deleteItem(); break;
```

```
        case 3: updateQuantity(); break;
```

```
        case 4: displayCart(); break;
```

```
        case 5: computeBill(); break;
```

```
        case 6: displayReverse(); break;
```

```
        case 7: exit(0);
```

```
        default: printf("Invalid choice.\n");
```

```
    }
```

```
}
```

```
return 0;
```

```
}
```

Output:

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Student\Desktop\25uam095> gcc microproject_25uam095.c
PS C:\Users\Student\Desktop\25uam095> ./a.exe

===== Shopping Cart Menu =====
1. Add Item
2. Delete Item
3. Update Quantity
4. Display Cart
5. Compute Bill
6. Display Cart in Reverse
7. Exit
=====
Enter your choice: 1
Enter Product ID: 101
Enter Product Name: Milk
Enter Unit Price: 35rs
Enter Quantity: 1
Product added to cart.

===== Shopping Cart Menu =====
1. Add Item
2. Delete Item
3. Update Quantity
4. Display Cart
5. Compute Bill
6. Display Cart in Reverse
7. Exit
=====
Enter your choice: 1
Enter Product ID: 102
Enter Product Name: Bread
Enter Unit Price: 30rs
Enter Quantity: 1
Product added to cart.
```

===== Shopping Cart Menu =====

1. Add Item
2. Delete Item
3. Update Quantity
4. Display Cart
5. Compute Bill
6. Display Cart in Reverse
7. Exit

=====

Enter your choice: 1
Enter Product ID: 103
Enter Product Name: cake
Enter Unit Price: 250rs
Enter Quantity: 1
Product added to cart.

===== Shopping Cart Menu =====

1. Add Item
2. Delete Item
3. Update Quantity
4. Display Cart
5. Compute Bill
6. Display Cart in Reverse
7. Exit

=====

Enter your choice: 1
Enter Product ID: 104
Enter Product Name: Mango
Enter Unit Price: 20rs
Enter Quantity: 6
Product added to cart.

===== Shopping Cart Menu =====

1. Add Item
2. Delete Item
3. Update Quantity
4. Display Cart
5. Compute Bill
6. Display Cart in Reverse
7. Exit

=====

Enter your choice: 4

Cart Items:

ID	Name	Price	Qty	Total
101	Milk	35.00	1	35.00
102	Bread	30.00	1	30.00
103	cake	250.00	1	250.00
104	Mango	20.00	6	120.00

===== Shopping Cart Menu =====

1. Add Item
2. Delete Item
3. Update Quantity
4. Display Cart
5. Compute Bill
6. Display Cart in Reverse
7. Exit

=====

Enter your choice: 4

Cart Items:

ID	Name	Price	Qty	Total
101	Milk	35.00	2	70.00
103	cake	250.00	1	250.00
104	Mango	20.00	6	120.00

===== Shopping Cart Menu =====

1. Add Item
2. Delete Item
3. Update Quantity
4. Display Cart
5. Compute Bill
6. Display Cart in Reverse
7. Exit

=====

Enter your choice: 5

Total Bill Amount: 440.00

===== Shopping Cart Menu =====

1. Add Item
2. Delete Item
3. Update Quantity
4. Display Cart
5. Compute Bill
6. Display Cart in Reverse
7. Exit

=====

Enter your choice: 6

Cart Items in Reverse:

104	Mango	20.00	6	120.00
103	cake	250.00	1	250.00
101	Milk	35.00	2	70.00

===== Shopping Cart Menu =====

1. Add Item
2. Delete Item
3. Update Quantity
4. Display Cart
5. Compute Bill
6. Display Cart in Reverse
7. Exit

=====

Enter your choice: 2

Enter Product ID to delete: 102

Product deleted.

===== Shopping Cart Menu =====

1. Add Item
2. Delete Item
3. Update Quantity
4. Display Cart
5. Compute Bill
6. Display Cart in Reverse
7. Exit

=====

Enter your choice: 4

Cart Items:

ID	Name	Price	Qty	Total
101	Milk	35.00	1	35.00
103	cake	250.00	1	250.00
104	Mango	20.00	6	120.00

===== Shopping Cart Menu =====

1. Add Item
2. Delete Item
3. Update Quantity
4. Display Cart
5. Compute Bill
6. Display Cart in Reverse
7. Exit

=====

Enter your choice: 3

Enter Product ID to update: 101

Enter new Quantity: 2

Quantity updated.

===== Shopping Cart Menu =====

1. Add Item
2. Delete Item
3. Update Quantity
4. Display Cart
5. Compute Bill
6. Display Cart in Reverse
7. Exit

=====

Enter your choice: 7

PS C:\Users\Student\Desktop\25uam095> |